

**Table 2: Final pharmacokinetic parameters estimate for isoniazid, rifampicin and pyrazinamide.**

| <b>Typical values (95% confidence interval) [Shrinkage]</b>  |                          |                         |                          |
|--------------------------------------------------------------|--------------------------|-------------------------|--------------------------|
| <b>Parameter</b>                                             | <b>Isoniazid</b>         | <b>Rifampicin</b>       | <b>Pyrazinamide</b>      |
| <b>Clearance (L/h)<sup>a</sup></b>                           |                          | 15.8 (13.5 – 18.7)      | 3.88 (3.54 – 4.08)       |
| Slow acetylator (L/h) <sup>a, c</sup>                        | 10.5 (8.97 – 11.3)       |                         |                          |
| Intermediate/Rapid acetylator (L/h) <sup>a, c</sup>          | 28.8 (24.3 – 29.5)       |                         |                          |
| <b>Population proportion of NAT2 acetylator groups</b>       |                          |                         |                          |
| Slow acetylators                                             | 0.305 (0.226 – 0.435)    |                         |                          |
| Fast/intermediate acetylators                                | 0.695 (0.565 – 0.774)    |                         |                          |
| <b>Central volume of distribution (L)<sup>a</sup></b>        | 41.3 (37.7 – 45.0)       | 50.8 (42.9 – 60.4)      | 40.1 (37.7 – 41.6)       |
| <b>Intercompartmental clearance (L/h)<sup>a</sup></b>        | 2.97 (2.13 – 3.65)       |                         |                          |
| <b>Peripheral volume of distribution (L)<sup>a</sup></b>     | 30.2 (22.1 – 43.1)       |                         |                          |
| <b>Bioavailability</b>                                       | 1 FIXED                  | 1 FIXED                 | 1 FIXED                  |
| <b>Number of absorption transit compartments</b>             |                          |                         | 4 FIXED                  |
| <b>Mean Transit time (h<sup>-1</sup>)</b>                    |                          |                         | 0.424 (0.0638 – 0.511)   |
| <b>First-order absorption rate constant (h<sup>-1</sup>)</b> | 3.14 (2.55 – 5.16)       | 1.03 (0.774 – 1.48)     | 3.45 (2.56 – 4.15)       |
| <b>Hepatic blood flow (L/h)<sup>a, b</sup></b>               | 76.3 FIXED               |                         |                          |
| <b>Metformin effect size %</b>                               |                          |                         |                          |
| Bioavailability                                              | -14.3 (-4.45 – -26.4)    | -24.0 (-7.83 – -36.3)   |                          |
| <b>Between Subject Variability (%)</b>                       |                          |                         |                          |
| Clearance                                                    | 16.9 (13.4 – 20.9) [14%] | 42.2 (34.1 – 53.6) [9%] | 31.4 (26.2 – 38.0) [10%] |

| <b>Between Occasion Variability (%)</b>   |                          |                          |                          |
|-------------------------------------------|--------------------------|--------------------------|--------------------------|
| Bioavailability - F                       | 21.8 (13.3 – 33.6) [34%] | 31.2 (22.4 – 42.7) [16%] | 16.9 (13.5 – 21.2) [11%] |
| First-order absorption rate constant - Ka | 156 (155 – 598) [14%]    | 31.3 (6.93 – 56.6) [41%] | 96.1 (78.3 – 348) [7%]   |
| Mean Transit time - MTT                   |                          |                          | 267 (126 – 408) [7%]     |
| <b>Residual unexplained error</b>         |                          |                          |                          |
| Proportional error (%)                    | 17.9 (15.7 – 21.2)       | 32.1 (29.0 – 37.8)       | 7.40 (5.72 – 8.90)       |
| Additive error (mg/L)                     | 0.02 FIXED <sup>c</sup>  | 0.02 FIXED <sup>c</sup>  | 0.935 (0.608 – 1.34)     |

<sup>a</sup> All the disposition parameters were allometrically scaled. The reported typical values refer to the typical individual in the cohort with a fat-free mass of 45 kg.

<sup>b</sup> Hepatic blood flow is described in the literature as 90 L/h for a standard 70 kg male with an FFM of 56 kg. The typical FFM in our cohort is 45 kg, and the hepatic blood flow was scaled accordingly.

<sup>c</sup> The reported value is apparent clearance which was calculated as  $CL_{intrinsic} \times F_u$  (fraction unbound = 0.95)